Zhiheng Jiang

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Education

University of California, Los Angeles 2024-2028
Bachelor of Science in Computer Science and Engineering (GPA: 4.00 / 4.00) *Los Angeles, CA*

- **Courses:** Machine Learning, Linear Algebra, Probability and Statistics, Multivariable Calculus, Data Structures and Algorithms, Operating Systems, Digital Design (Verilog), Systems and Signals, GIS
- **Awards:** 2025 UCLA Samueli Undergraduate Scholarship Recipient, Dean's Honors List

Experience

UCLA Computational Machine Learning Lab Oct 2025 – Present
Undergraduate Researcher *Los Angeles, CA*

- Working under Prof. Cho-Jui Hsieh to develop a capability-preserving LLM activation steering method for truthfulness and safety
- Designed inference-time activation steering hooks for vLLM, and ran latency evaluations for activation steering methods
- Trained linear activation probes with logistic regression to investigate separation of concepts in the LLM activation space

Institute for Creative Technologies, University of Southern California June 2025 – Sep 2025
Visiting Academic *Playa Vista, CA*

- Researched on team-based agents with single meta-RL policy for U.S. DEVCOM Army Research Lab, advised by Dr. Volkan Ustun
- Designed benchmark problems for Preference-Driven Multi-Objective Reinforcement Learning with novel evaluation metrics
- Designed Heterogeneous Graph Neural Network with flexible preference conditioning for Proximal Policy Optimization (PPO)
- Authored a Python toolkit (GraphAllocBench) using Gymnasium, Stable Baselines3, PyTorch Geometric, Wandb and PyMOO
- Accepted first-author paper at the IJCAI-ECAI Multi-Objective Decision Making (MODeM) workshop in Bremen, Germany

Structures-Computer Interaction Lab at UCLA Oct 2024 – June 2025
Undergraduate Researcher *Los Angeles, CA*

- Developed digital twin simulation of springs using Large Language Models and Robot Tool Calling, advised by Prof. Khalid Jawed
- Refined numerical optimization methods using PyTorch Differential Equations (torchdiffeq) for reduced-order spring simulations
- Trained Physics-Informed Neural Networks with Autograd and Discrete Differential Geometry to predict spring elastic energy
- Worked with Amazon AWS to develop LangGraph clone from scratch to manage LLM multi-agent LLM memory with RAG
- Developed LLM tools for C++ OpenGL physics simulations, Intel RealSense video input (OpenCV) and Sawyer Robots (ROS)

Institute of High Performance Computing, A*STAR June 2020 – Feb 2022
Research Assistant *Singapore*

- First-author journal publication on large social network analysis with Dr. Hoai Nguyen Huynh, cited by Spotify Research (2024)
- Performed K-means, Girvan-Newman and Shannon Entropy clustering on scale-free user-oriented networks with NetworkX
- Improved community detection algorithms on large social networks parsed with BeautifulSoup from an AJAX review database
- Performed text mining using Natural Language Tool Kit (NLTK), TF-IDF and dependency parsing to describe genre communities

Publications

[1] **Zhiheng Jiang**, Yunzhe Wang, Ryan Marr, Ellen Novoseller, Benjamin T. Files and Volkan Ustun, "GraphAllocBench: A Flexible Benchmark for Preference-Conditioned Multi-Objective Policy Learning," *Proceedings of the IJCAI 2026 Workshop on Multi-Objective Decision Making (MODeM 2026)*, 2026.

[2] **Zhiheng Jiang** and Hoai Nguyen Huynh, "Unveiling music genre structure through common-interest communities" *Social Network Analysis and Mining*, Vol. 12, No. 35 (2022).

Projects

FreeClawRouter | *OpenClaw, Docker, Ollama, Nvidia NIM, OpenRouter, FastAPI* 2026

- Designed a reverse proxy for OpenClaw, using smaller on-device LLMs to make OS-like scheduling decisions for deploying larger cloud-based LLMs based on cross-platform usage statistics, under varying provider-dependent rate limit constraints
- Implemented an off-the-critical-path summarization engine that compresses conversational memory to maintain context limits

Robot Object Detection via Verbal Commands (AI Hackathon) | *LLMs, VLMs, Finetuning, PyTorch, HuggingFace, VertexAI, Docker* 2024

- AI Hackathon Champion, winning 10,000 SGD (7,500 USD) cash prize competing against 60 university-level finalist teams
- Finetuned large deep learning models with high test scores, for audio (99.5%), Vision-Language Models and object detection (86.3%) and Transformer Question Answering (99.9%), with quantization, to achieve high inference speeds on a DJI robot
- Programmed a robot to respond to verbal commands (audio and NLP), and identify targets (VLM, Object Detection) in 3D world
- Finetuned SOTA models such as YOLO, DETR, RoBERTa and OpenAI Whisper on VertexAI and Google Cloud Platform (GCP)

Technical Skills

Languages: Python, C++, MATLAB, Java, Javascript, SQL, LaTeX, Shell Scripting (Bash), YAML, JSON, HTML, Verilog
Technologies: PyTorch, HuggingFace, LangGraph, LangChain, LlamaIndex, Pandas, Numpy, ReactJS, NodeJS, FastAPI, Claude Code
Concepts: Large Language Models, Machine Learning, Computer Vision, Natural Language Processing, Reinforcement Learning